Exno: 8
Date: 20.8.26

NMAP to Discover live Host Using NMap Scans (ARP, ICMP, TCP (UDP) on the Toy. Hack Me platform.

Aim:

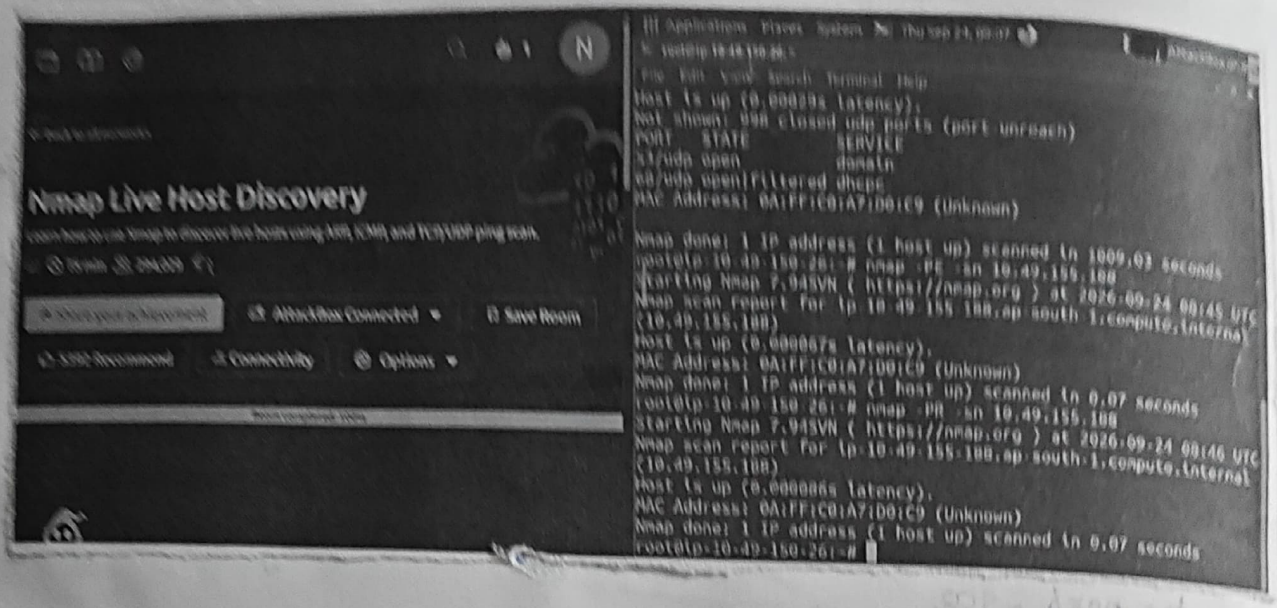
To Study the working of the Nmap tool and discover live hosts in a network using different host discovery Techniques such as ARP Scanning, ICMP Scanning, TCP and UDP Scanning on the TryHackMe platform.

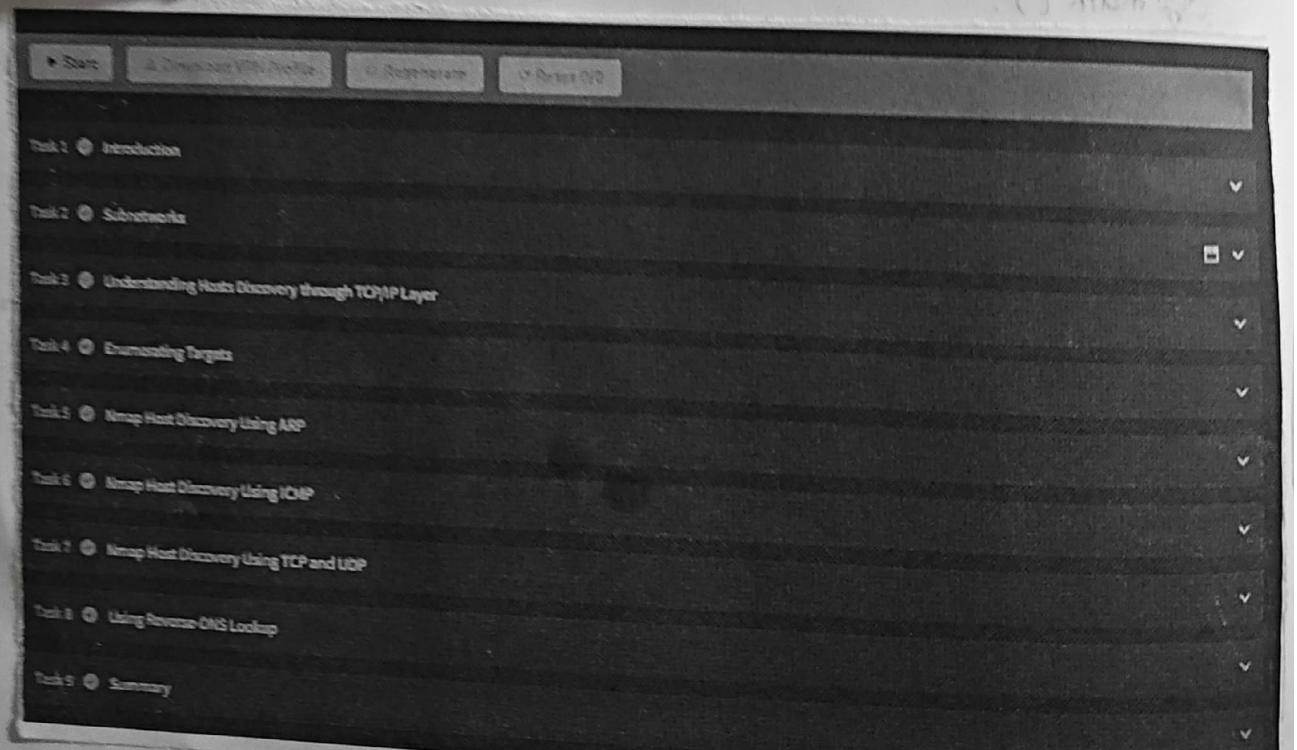
Step:

1) Open the TryHackMe platform and deleete The NMap room
2) Study the network topology & identify the source and destination hosts.
3) Use ARP requeles to discover the MAC Addres for local network.
4) Observe how ARP packets are broadcast with the network and identify the devices the receive the request.
5) Use the Icmp Protocol to Send ping request between two system and observe the response.
6) Analyse the packet flow and identify the exchange of ARP packets - before ICMP communication.
7) Perform ARP host detect using the Nmap command.
8) perform ICMP Echo Scanning and observe the host that respond to the Scan.
9) Perform ICMP Timestamp Scanning and Analyse them by target system.
10) perform ICMP Address mask Scanning and Compare the result with the previous scans.
11) perform TCP Syn Scanning to identify live hosts by analyzing SYN and ACK packet
12) perform TCP Ack scanning to verify.
13) perform TCP Ack Scanning to verify how Nmap identifies active hosts Through UDP Packets.
14) Perform a Reverse DNS lookups to obtain the hostname of the discovered system.
15) Compare the result obtained from the ARP, ICMP, TCP and UDP scanning.

Result:

Thus, the hosts in the network was successfully discovered using ARP, ICMP, TCP, or UDP scanning techniques with the Nmap tool.


Nmap Live Host Discovery
Learn how to use Nmap to discover live hosts using ARP, ICMP, and TCP/UDP ping scan.
AttackBox Connected
Save Room
Connectivity
Options
Applications Places System
Thu Sep 24, 00:07
File Edit View Search Terminal Help
Host is up (0.00029s latency).
Not shown: 998 closed udp ports (port unreach)
PORT STATE SERVICE
53/udp open domain
68/udp open|filtered dhcpc
MAC Address: 0A:FF:C0:A7:D0:C9 (Unknown)
Nmap done: 1 IP address (1 host up) scanned in 1009.03 seconds
root@ip-10-49-150-26:~# nmap -PE -sn 10.49.155.188
Starting Nmap 7.94SVN (https://nmap.org) at 2026-09-24 00:45 UTC
Nmap scan report for ip-10-49-155-188.ap-south-1.compute.internal
(10.49.155.188)
Host is up (0.00067s latency).
MAC Address: 0A:FF:C0:A7:D0:C9 (Unknown)
Nmap done: 1 IP address (1 host up) scanned in 0.07 seconds
root@ip-10-49-150-26:~# nmap -PR -sn 10.49.155.188
Starting Nmap 7.94SVN (https://nmap.org) at 2026-09-24 00:46 UTC
Nmap scan report for ip-10-49-155-188.ap-south-1.compute.internal
(10.49.155.188)
Host is up (0.00008s latency).
MAC Address: 0A:FF:C0:A7:D0:C9 (Unknown)
Nmap done: 1 IP address (1 host up) scanned in 0.07 seconds
root@ip-10-49-150-26:~#


Start
Task 1 Introduction
Task 2 Subnetworks
Task 3 Understanding Hosts Discovery through TCP/IP Layer
Task 4 Enumerating Targets
Task 5 Nmap Host Discovery Using ARP
Task 6 Nmap Host Discovery Using ICMP
Task 7 Nmap Host Discovery Using TCP and UDP
Task 8 Using Reverse-DNS Lookup
Task 9 Summary